

Suman Majumder

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Education

Aug 2016 – Jun 2020 Ph.D., Statistics, North Carolina State University

Thesis Title: Spatiotemporal inference and applications
for large datasets

Advisor: Brian J. Reich

Jul 2014 – May 2016 M.Stat., Indian Statistical Institute, Kolkata

Jul 2011 – May 2014 B.Stat. (Hons.), Indian Statistical Institute, Kolkata

Experience

Dec 2024 – Current Assistant Professor, Interdisciplinary Statistical Research Unit,
Indian Statistical Institute, Kolkata

Aug 2023 – Dec 2024 Assistant Professor, Department of Statistics, University of
Missouri

Jul 2020 – Jun 2023 Postdoctoral Fellow, Department of Nutrition, Harvard T. H.
Chan School of Public Health

Jan 2020 – Jun 2020 Research Assistant, Department of Entomology and Plant
Pathology, North Carolina State University

Aug 2018 – Dec 2019 Teaching Assistant, Department of Statistics, North Carolina
State University

Aug 2017 – Jun 2018 Research Assistant, The Statistical and Applied Mathematical
Sciences Institute (SAMSI)

Aug 2016 – May 2017 Teaching Assistant, Department of Statistics, North Carolina
State University

Teaching

Courses

- Indian Statistical Institute
 - Statistical Data Analysis, Spring 2026; Statistical Theory and Applications 77th Term (International Statistical Education Centre)
 - Statistics Comprehensive, Spring 2026; B. Stat.(Hons.) (Jointly with Dr. Debasis Sengupta)
 - Inference, Fall 2025; PGDBA
 - Basic Statistics, Spring 2025, Fall 2025, Spring 2026; PGDAS (Coursera)
 - Sample Surveys and Design of Experiment, Spring 2025; M. Stat. (Jointly with Dr. Kajal Dihidar)
- University of Missouri
 - STAT 4760/7760: Statistical Inference, Fall 2024, 2023
 - STAT 9370: Multivariate Analysis, Spring 2024

Guest Lectures

- Harvard School of Public Health
 - BST 228: Applied Bayesian Analysis, Fall 2021
- North Carolina State University
 - ST 422: Introduction to Mathematical Statistics, Spring 2019
 - ST 740: Introduction to Bayesian Inference, Fall 2018

Honors and Awards

- 2026: Professor Aditya Chattopadhyay Memorial Young Scientist Award, Calcutta Statistical Association
- 2021: Richard L. Anderson Award at SRCOS Summer Research Conference
- 2021: Boyd Hashbarger Travel Award for SRCOS Summer Research Conference
- 2020: ASA ENVR Student Paper Award Winner
- 2019: Travel Award for IMS/ASA Spring Research Conference
- 2018 - current: Member of NC Alpha chapter of Statistics, *Mu Sigma Rho*
- 2011 - 2016: KVPY Scholarship, Dept. of Science & Technology, Govt. of India

Publications (*:co-first author)

Peer-Reviewed Journals

- Lee, K. H., Coull, B. A., **Majumder, S.**, La Riviere, P. J., Mark Welch, J. L., Starr, J. R.; A Bayesian Multivariate Spatial Point Pattern Model: Application to Oral Microbiome FISH Image Data; *Accepted, Biostatistics*
- **Majumder, S.**, Sahoo, I. (2026); [Bayesian Variable Selection for Censored Spatial Responses with Application to PFAS Concentrations in California](#). Spatial Statistics
- Sahoo, I., **Majumder, S.***, Hazra, A., Rappold, A. G., Bandyopadhyay, D. (2025); [Computationally Scalable Bayesian SPDE Modeling for Censored Spatial Responses](#). New England Journal of Statistics and Data Science
- **Majumder, S.**, Coull, B. A., Mark Welch, J. L., La Riviere, P. J., Dewhurst, F. E., Starr, J. R., Lee, K. H. (2024); [Multivariate cluster point process to quantify and explore multi-entity configurations: Application to biofilm image data](#). Statistics in Medicine
- Giffin, A., Gong, W., **Majumder, S.**, Rappold, A. G., Reich, B. J., Yang, S. (2022); [Estimating intervention effects on infectious disease control: the effect of community mobility reduction on Coronavirus spread](#). Spatial Statistics.
- **Majumder, S.**, Guan, Y., Reich, B. J., Saibaba, A. K. (2022); [Kryging: Geostatistical analysis of large-scale datasets using Krylov subspace methods](#). Statistics and Computing.
- Dorman, S. J., Hopperstad, K. A., Reich, B. J., **Majumder, S.**, Kennedy, G., Reisig, D. D., Greene, J. K., Reay-Jones, F. P. F., Collins, G., Bacheler, J. S., and Huseth, A. S. (2021); [Landscape-level variation in *Bt* crops predict *Helicoverpa zea* \(Lepidoptera: Noctuidae\) resistance in cotton agroecosystems](#). Pest Management Science.
- **Majumder, S.**, Guan, Y., Reich, B. J., O'Neill, S. and Rappold, A. G. (2020); [Statistical Downscaling with Spatial Misalignment: Application to wildland Fire PM_{2.5} Concentration Forecasting](#). Journal of Agricultural, Biological and Environmental Statistics.
- **Majumder, S.**, Biswas, A., Roy, T., Bhandari, S., and Basu, A. (2020); [Statistical Inference Based on a New Weighted Likelihood Approach](#). Metrika.
- Haskett, M. E., **Majumder, S.**, Kotter-Grühn, D. and Gutierrez, I. (2020); [The role of university students' wellness in links between homelessness, food security, and academic success](#). Journal of Social Distress and the Homelessness.
- Biswas, A., **Majumder, S.**, Niyogi, P. G. and Basu, A. (2020); [A Weighted Likelihood Approach to Problems in Survival Data](#). Sankhyā.
- Haskett, M. E., Kotter-Grühn, D. and **Majumder, S.** (2020); [Prevalence and correlates of homelessness and food insecurity among university students](#). Journal of College Student Development.
- Biswas, A., Roy, T., **Majumder, S.**, and Basu, A. (2015). [A new weighted likelihood approach](#). Stat.

Under Review/ Submitted

- **Majumder, S.**, Chakraborty, S., Shyu, C. R., Raghavan, R. K.: CensGSTCAR: A Bayesian Model for Censored Spatiotemporal Count Data with Application to Tick-Borne Disease Surveillance; *Submitted, Spatial Statistics*
- **Majumder, S.**, Jun, Y., Chakraborty, S., Lim, C. Y., Dey, T.: BSTZINB: A Bayesian Framework for Spatio-Temporal Modeling of Zero-Inflated Count Data in Epidemiology; *Submitted, Journal of Data Science*

In Preparation

- Wang, S., **Majumder, S.***, Coull, B. A., Mark Welch, J. L., La Riviere, P. J., Starr, J. R., Lee, K. H.: A Generalized Bayesian Framework for Multi-Level Spatial Point Patterns via Multivariate Log-Gaussian Cox Processes (2025+)
- Gong, Y., **Majumder, S.**, Coull, B. A., Mark Welch, J. L., La Riviere, P. J., Starr, J. R., Lee, K. H.: Multivariate spatial LGCP modeling with INLA-SPDE, with application to human microbiome imaging data (2025+)
- Ghosal, R., **Majumder, S.***, Sahoo, I.: Fast spatially indexed longitudinal distributional outcome regression for predicting monthly distributional representations of total nitrate (TNO_3) concentrations in the USA (2025+)
- **Majumder, S.**, Coull, B. A., Mark Welch, J. L., Starr, J. R., Lee, K. H.: A Bayesian approximate multivariate log-Gaussian Cox process model to approximate multiple 2D z-slices of a 3D image (2025+)
- Bater, J., **Majumder, S.***, Huang, Y., Moscicki, A., Yao, T., Lee, K. H., Starr, J. R.: Oral pathogen Filifactor alocis's relative abundance is lower among HIV-positive youth with less antiretroviral therapy exposure (2025+)

Technical reports

- **Majumder, S.**, Gopi, K., Horsley, J., Yuen, C., Cope, M., Morgan, G. (2023). Enhanced AQVx Near Real Time Blending Utility.
- Biswas, A., Niyogi, P. G., **Majumder, S.**, Bhandari, S. K., Ghosh, A., and Basu, A. (2018). [Theoretical Properties of a New Weighted Likelihood Estimator for Right Censored Data.](#)

Software

- BSTZINB [R package, [CRAN](#)]
- MCPP [R package, [GitHub](#)]
- Kryging [Matlab, [GitHub](#)]
- Warping utility [R, [GitHub](#)]

Mentoring

Ph.D. Advisor/Co-advisor

- Vivek Singh (Ph.D., Statistics, Expected 2029), Indian Statistical Institute

Master's Degree Thesis Advisor/Co-Advisor (*: Dissertation)

- Archishman Mukherjee (M.Stat., Statistics, 2026), Indian Statistical Institute [Jointly with Debashis Paul]
- Tanmoy Nath (M.Stat., Statistics, 2026), Indian Statistical Institute
- Ramanuj Das (M.Stat., Statistics, 2026), Indian Statistical Institute
- Simanto Mukherjee (M.Stat., Statistics, 2026), Indian Statistical Institute
- Sayak Panja* (M.Stat., Statistics, 2026), Indian Statistical Institute
- Kushal Sedhai (M.A., Statistics, 2024), University of Missouri

Ph.D. Graduate Committee Member/External

- Myungsoo Yoo (Ph.D., Statistics, 2024), University of Missouri

Presentations

Invited Talks

- 2026 IISA Conference, Benaras Hindu University, Varanasi, India, 2026
- 9th International Conference on Econometrics and Statistics (EcoSta 2026), Ryukoku University, Kyoto, Japan, 2026
- ISI-ISM-ISSAS Joint Conference 2026, Institute of Statistical Sciences Academia Sinica, Taipei City, Taiwan, 2026
- Statistical Sciences Unit, Indian Statistical Institute, Delhi Centre, New Delhi, India, 2025
- International Chinese Statistical Association Applied Statistics Symposium, Nashville, TN, USA, 2024
- International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2023), Berlin, Germany, 2023
- Department of Statistical Sciences and Operations Research, Virginia Commonwealth University, Richmond, VA, USA, 2023
- Department of Statistics 60th Anniversary Conference, University of Missouri, Columbia, MO, USA, 2023

- Department of Epidemiology and Biostatistics, Arnold School of Public Health, University of South Carolina, Columbia, SC, USA, 2023.
- International Chinese Statistical Association Applied Statistics Symposium, Gainesville, FL, USA, 2022.
- Department of Mathematics and Statistics, University of Massachusetts-Amherst, Amherst, MA, USA, 2022.
- Center for Air Pollution, Energy and Health Research, Webinar series, 2020.

Contributed Talks

- Joint Statistical Meetings, Washington DC, USA, 2022.
- ENAR 2021 Spring Meeting, Online, 2021.
- ENVR Student Paper Awards Section, Joint Statistical Meetings, Online, 2020.
- International Conference on Robust Statistics, Kolkata, West Bengal, India, 2015.

Posters

- ASA ENVR Workshop, Boulder, CO, USA, 2024
- ASA/Kutner Faculty Poster Session, SRCOS Summer Research Conference, Clemson, SC, USA, 2024
- Boyd Hashbarger Student Poster Competition, SRCOS Summer Research Conference, Jekyll Island, GA, USA, 2021.
- Statistical Significance poster competition, Joint Statistical Meetings, Denver, CO, USA, 2019.
- IMS/ASA Spring Research Conference, Blacksburg, VA, USA, 2019.
- SAMSI Graduate Poster Session, Raleigh, NC, USA, 2018.
- IISA International Conference on Statistics, Gainesville, FL, USA, 2018.

Funding

- Comprehensive Study on Machinery Footprint & Employment Scenarios in India's Textile & Apparel Sector. *Subject Matter Expert*. Amount Received: INR 24,72,500. Ministry of Textiles, Government of India. July 2026 - Dec 2026.
- Workshop on Spatial and Spatiotemporal Statistical Analysis. *Co-organizer*. Amount Received: INR 90,000. DCSW – StatSD, Indian Statistical Institute. Dec 2026.
- Start-up Grant. *Principal Investigator*. Amount Received: INR 2,75,000. Indian Statistical Institute. Aug 2025 - Mar 2026.

Professional Service

As Reviewer or Editor

- Served as Reviewer for:
 - **Statistical Journals:** Statistics and Computing, Journal of the Royal Statistical Society Series B, Journal of the Royal Statistical Society Series C, Statistics in Medicine, Spatial Statistics, Sankhya Series B, Journal of Agricultural, Biological and Environmental Statistics, Computational Statistics and Data Analysis, Communication in Statistics - Simulation and Computation
 - **General Scientific Journals:** Public Library of Sciences ONE, Environmental and Ecological Statistics, Ecological Applications, Econometrics and Statistics, Springer Nature Applied Sciences
- Served as Editor for:
 - Technical Editor for Sankhya: Series B

Judging/Paneling

- Judge for the Mizzou DataFest in 2024

Conferences/Workshops Organized

- RobStat Symposium 2027, 2027, Kolkata, WB, IN
- Basu-Bahadur Memorial Conference 2026, 2026, Kolkata, WB, IN
- Workshop on Spatial and Spatiotemporal Statistical Analysis, 2026, New Delhi, DL, IN
- Workshop on Statistical Computing, 2026, New Delhi, DL, IN
- Workshop on Applied Spatial Statistics, 2026, Virtual
- Half-day Workshop on Modern Approaches to Analyzing Complex Data, 2025, Kolkata, WB, IN

Conference Sessions Organized

- Invited Session, International Indian Statistical Association Conference, 2026, Varanasi, UP, IN (*jointly with Dr. Kaushik Jana, ISI Delhi*)

Memberships

- International Indian Statistical Association, *Life Member*
- International Statistical Institute, *Member*

Social Activities

- Co-organizer, 20 Year Anniversary of ISRU (Formerly BIRU), Interdisciplinary Statistical Research Unit, Indian Statistical Institute Kolkata, 2026
- Organizer, Departmental Diwali Celebration, Department of Statistics, University of Missouri, 2024, 2023.
- Volunteer, Service Raleigh, 2019.
- Head, Sponsorship Committee, Integration: the annual techno-cultural fest of Indian Statistical Institute, Kolkata, 2015, 2014.
- Member, Literary Activities Committee, Boys' Hostel of Indian Statistical Institute, 2014, 2013, 2012.
- Head, Advertising Committee, Reunion Celebration Committee of Rahara Ramakrishna Mission Boys' Home, 2013.